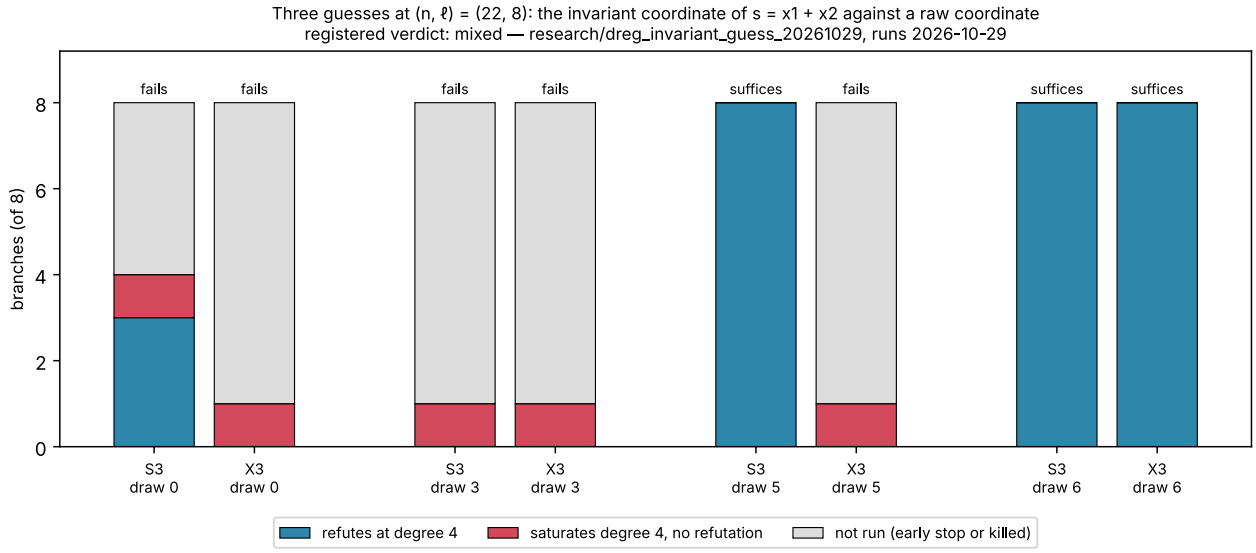
An invariant guess is never worse than a raw guess at $\ell = 8$, but does not reliably suffice

Study `research/dreg_invariant_guess_20261029/`, pre-registered. **Class:** stage diagnostic. **Registered verdict:** mixed.

Question

The first link of the m = 3 chained S_3 system sees its two summands only through the swap-invariants $s = x_1 + x_2$ and $p = x_1 x_2$. At an equal count of three guesses, does fixing the invariant coordinate $a_7 + b_7$ of s let the degree-4 MutantXL closure refute where fixing the raw a_7 does not?

Result



draw	S3 (invariant)	X3 (raw)
0	3 refute, then a failure	fails at branch 0
3	fails at branch 0	fails at branch 0
5	**all 8 refute**	fails at branch 0
6	all 8 refute	all 8 refute

- **Registered verdict:** mixed. S3 suffices on 2 of 4 draws, a split; X3 suffices on 1 of 4.
- **Paired by draw:** S3 is never worse, and is strictly better on draw 5.
- **Controls valid:**
  - the instrument identity and positive controls pass;
  - soundness, under a post-hoc rule (Amendment 1): every refutation of the satisfiable draw is in a branch with no solution, checked against the full solution list.

Reading

- It is the first measured guess choice that beats raw coordinates at equal k on the same draws, but on only 1 to 2 of 4 draws. That is a direction, not an established effect.
- It weakens the summand-dimension reading: S3 leaves both summands at dimension 8 and still refutes on two draws.
- No row moves and none beats rho. The guess count at the priced cell stays far above the oracle's break-even of 6.
- **Successors:** more draws at this cell; two invariant guesses on the pair; m = 4.